



The Value of Data

Lecture #19

The concept of value of an asset

- There is a well defined concept of value of all the conventional types of assets that we are familiar with
 - Oil, water, currency, infrastructure etc.
 - Unit of measure is well defined
 - Value is well defined for a unit measure and total value is a simple multiplication of unit measure value and total quantity
- In the case of data
 - There is no universally applicable unit measure
 - There is no simple way to assign a “value” to either a unit measure or a given quantity of data

Challenges in ascribing value to data

- Cost of acquiring?
 - Are all inputs costed? – eg. Clickstream data collection – are users compensated?
 - When copies are made does each copy diminish in value and hence does the entire collection diminish in value
- Potential to monetize?
 - Outright selling of data?
 - How does the value of one copy of the data permit the computation of the data overall
 - The incremental cost of a copy is negligible in contrast with conventional assets
 - Value realized through specific business models?
- Value at risk?
 - The costs of dealing with data destruction/damage/misuse
- Inherent value
 - Can data be used as a reliable and context independent value store?

Current Asset Accounting Practices

- Many (most?) data assets fail to meet the recognition criteria under conventional accounting standards which are
 - Identifiability
 - Control
 - Future economic benefits
 - Reliable measurement
- Consequently the accounting treatment of data
 - Expenses the costs of data collection and processing
 - Qualitative disclosure of data
 - Indirect valuation

Consequences of weak data valuation methods

- Inability to correctly value an asset results in
 - Inability to justify the money and resources to maintain the asset
 - Inability to justify the money and resources to secure the asset
- Inability to carry out a realistic risk assessment in regard to the value of the asset and the value at risk if the asset is impaired
- Inability to transfer risk by way of obtaining insurance

Cost of acquisition based models

- Costs of collection of data
- Cost of storage and infrastructure
- Cost of quality assurance
- Cost of security measures
- Cost of compliance testing and reporting
- Cost of maintaining and updating the data

Models based on ability to monetize

- A category of models exists based on the ability to generate income from the data
 - Direct sale of the data
 - Purchase a copy of the data
 - Value based on estimated income over a period
 - Leverage data in a service based business model
 - Google ads where personal data is monetized by creating a targeted advertising service where advertisers pay based on the number of clicks their ads receive
 - Value is again based on estimated revenue over a period of time
- These models are more context dependent than the cost based models since different organizations may have widely differing capabilities to monetize the data

Models based value at risk

- Models based on value at risk involve
 - Direct costs associated with data destruction/degradation
 - Restoration costs
 - Fines and penalties
 - Indirect costs associated with the data destruction/degradation
 - Loss of opportunity
 - Claims by third parties for costs inflicted on them

USE OF DATA TO STORE VALUE

Money

- **Commodity money**

Money whose value comes from a commodity of which it is made. Commodity money consists of objects having value or use in themselves (intrinsic value) as well as their value in buying goods

- **Representative money**

Money which has no intrinsic value but represents something of value such as gold or silver, in which it can be exchanged through a formal process

- **Fiat money**

Money which derives its value from having been established as money by government regulation

Digital Currency

- Money-like asset stored and exchanged on digital computers most often over the internet but may also be in devices like smartcards
- Can be
 - centralized – Money supply is managed by a central entity – usually the Central Bank of a country
 - Decentralized – decided democratically and not controlled by any single entity - Bitcoin

Digital v/s Virtual Currency

- Virtual currency is defined by the US Department of Treasury as
 - *a medium of exchange that operates like a currency in some environments, but does not have all the attributes of real currency*
 - *Virtual currency does not have legal-tender status in any jurisdiction*
- According to the European Central Bank
 - *virtual currency is a digital representation of value, not issued by a central bank, credit institution or e-money institution, which, in some circumstances, can be used as an alternative to money*

Bank Money

- A very high percentage of all money is in accounts in Banks
- This money is stored in digital form on digital computer systems
- Therefore it has been argued that this is also a form of Digital Currency

Digital v/s Cryptocurrency

- Cryptocurrency is a form of value storage where the generation of value and records of its transaction are maintained through the use of cryptographic methods. It is generally but not always the case that such systems are decentralized
- In decentralized systems value supply is controlled through
 - Proof of work
 - Proof of stake
- The oldest and still most popular system is Bitcoin
- A central bank might also use a distributed ledger technology to maintain an immutable record of transactions and control money supply centrally. In such cases the term cryptocurrency refers to Central Bank issued Digital Currency
- In the case of distributed systems of cryptocurrency such as bitcoin, these are often classified as virtual currency

Origin of the term blockchain

- The term “blockchain” is composed of two terms “block” and “chain” which appeared as separate terms in the seminal paper on bitcoin¹
- A “block” is a cryptographically secured collection of data. Various “blocks” are linked together to form a “chain”
- Somewhere down the line the technology has become known as “blockchain” technology

1. Nakamoto, Satoshi, “Bitcoin: A Peer-to-Peer Electronic Cash System” ,(31 October 2008).

History – The runup to bitcoin

- Chaum David Lee, Computer Systems Established, Maintained and Trusted by Mutually Suspicious Groups – Ph.D. dissertation, UC Berkley, 1982
 - Introduced the problem formulation of creating trustworthy computer systems that are trusted by parties who do not trust each other
 - Cryptographic techniques used to make data trustworthy

History – The runup to bitcoin

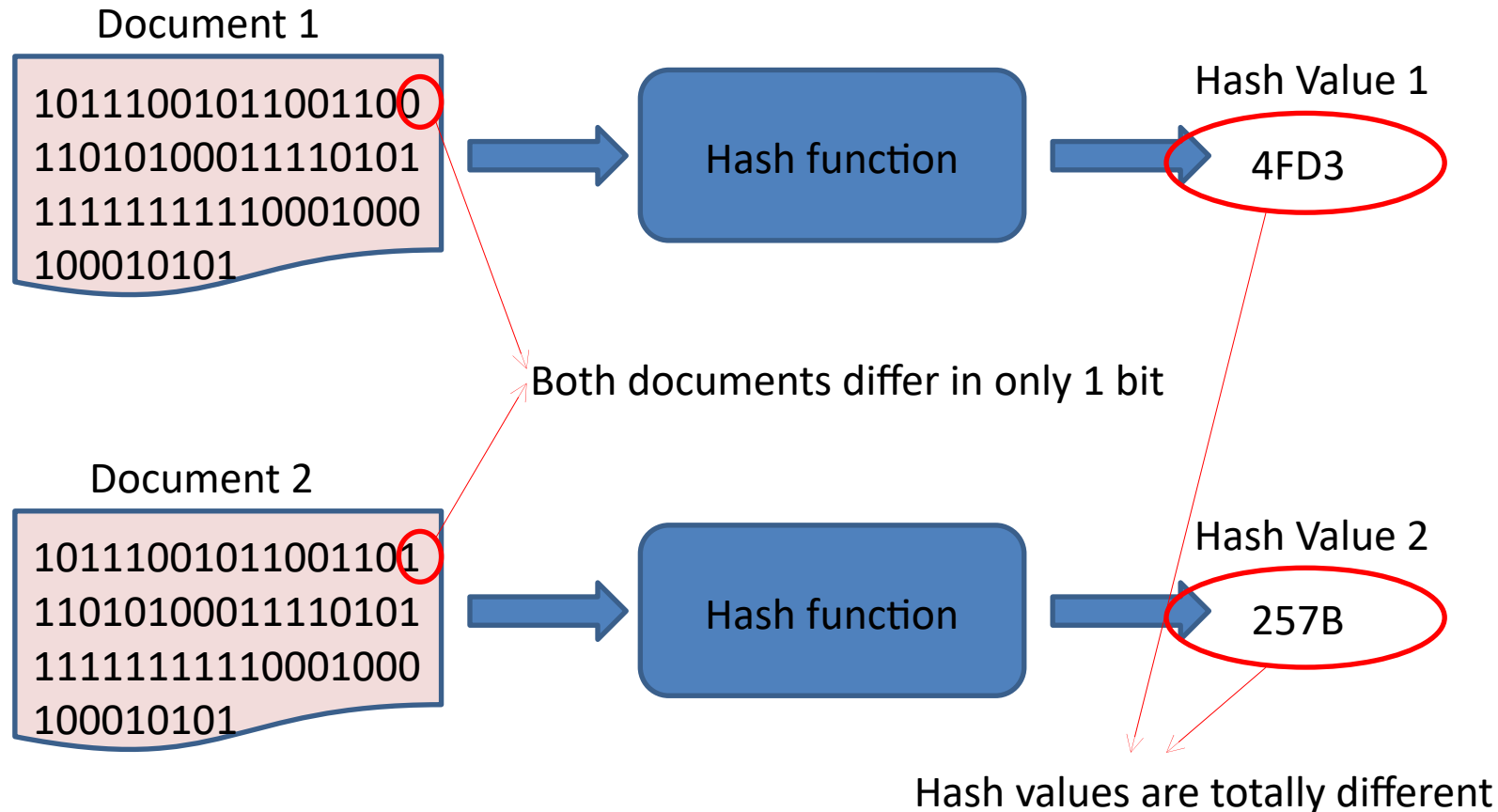
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Technological Precursors

- In 1991¹ Haber *et al* developed a technique to ensure that the timestamp on a digital document could not be tampered with
- In 1992² Bayer et al improved the efficiency of the technique

1. Haber, Stuart; Stornetta, W. Scott (January 1991). "How to time-stamp a digital document". *Journal of Cryptology*. **3** (2): 99–111.
2. Bayer, Dave; Haber, Stuart; Stornetta, W. Scott (March 1992). *Improving the Efficiency and Reliability of Digital Time-Stamping. Sequences*. **2**. pp. 329–334.

Hash Functions



Modern, well designed hash functions ensure that a change in the documents cannot be correlated to a change in the hash values. Even a single bit being different can produces radically different hash values

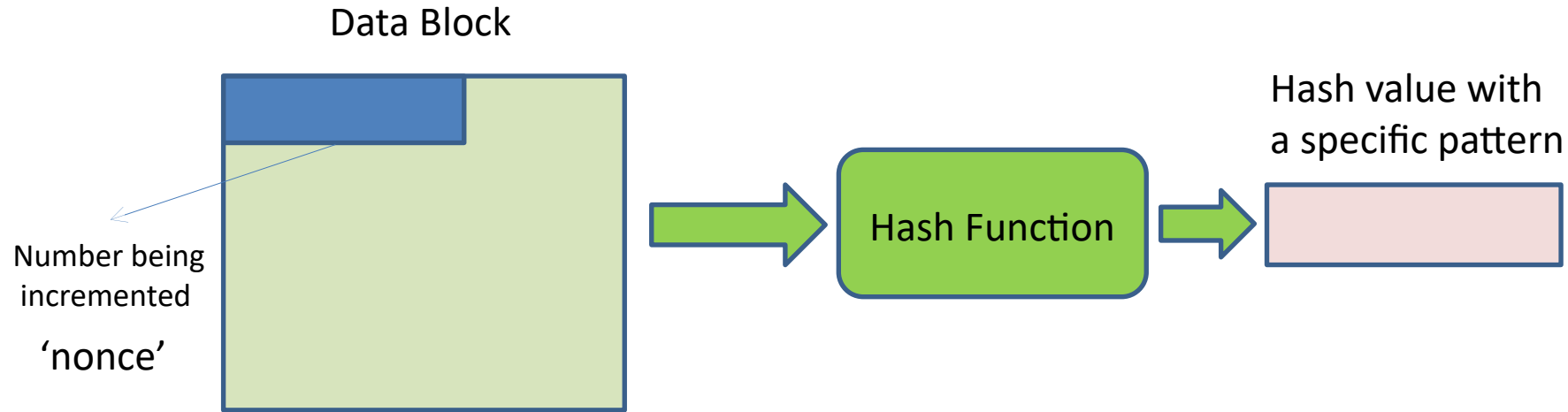
Hash Functions

- A hash function takes in an arbitrary input and produces a (usually) small fixed size output
 - An example of a (bad) hash function is the number of vowels in a document
- A hash function can be said to “summarize” a document
- A good hash function will produce uncorrelated hash values for correlated documents
- Since the set of all possible hash values is much smaller than the set of all possible documents, it is possible that many distinct document may have the same hash value. This is called a collision.
- It is extremely unlikely that two meaningfully correlated documents have a collision. Therefore it is safe to use a well designed hash function as a summary for a document
- SHA-256 is an example of a modern well defined hash function

Proof of work

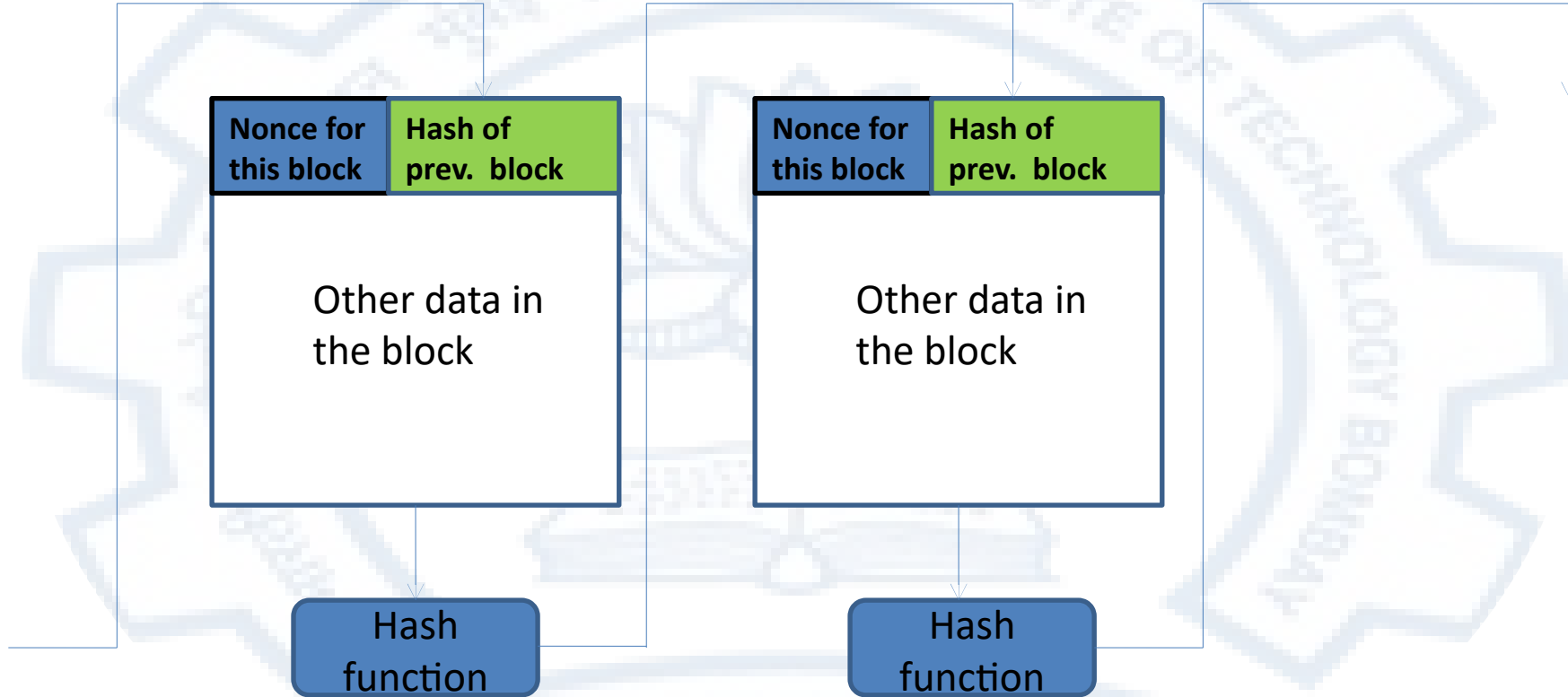
- Proof of work is a computation that a in which the party “that has done the work (prover)” establishes to any other parties (verifiers) that a significant but not infeasible amount of computational power has been spent
- The verifiers can all verify that the work has been done by carrying out a simple computation

An example of proof of work



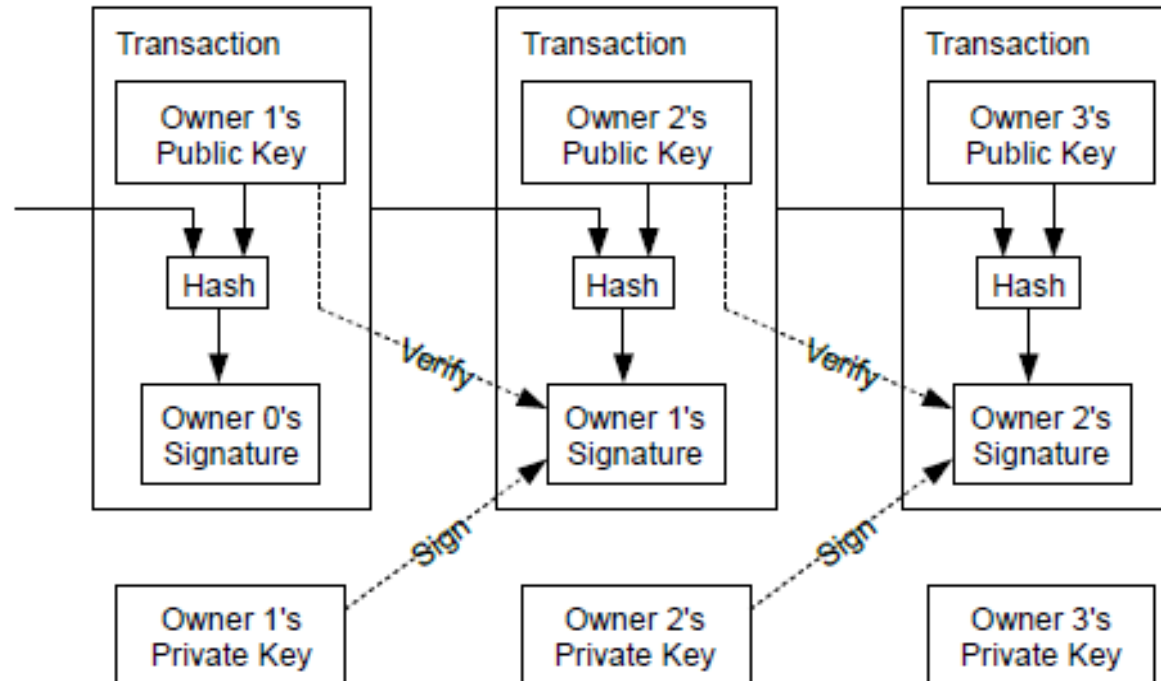
- A well designed hash function produces uncorrelated hash values when given correlated inputs
- The prover takes a given data block that has a specific placeholder for a number. The prover can change this number but nothing else in the data block.
- The prover must chose a number such that the resultant hash function satisfies some condition. For bitcoin this condition is that the hash must start with a number of zeros. Currently the number is 18 zeros
- The prover must do some significant work
- A verifier can verify by a single hash computation

Blockchain



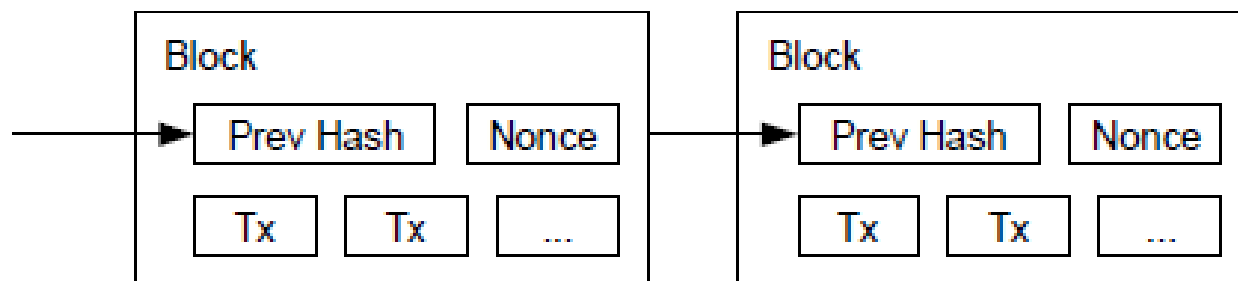
A change in any block requires all subsequent blocks to be recomputed so that their hashes conform to the requirement

Bitcoin

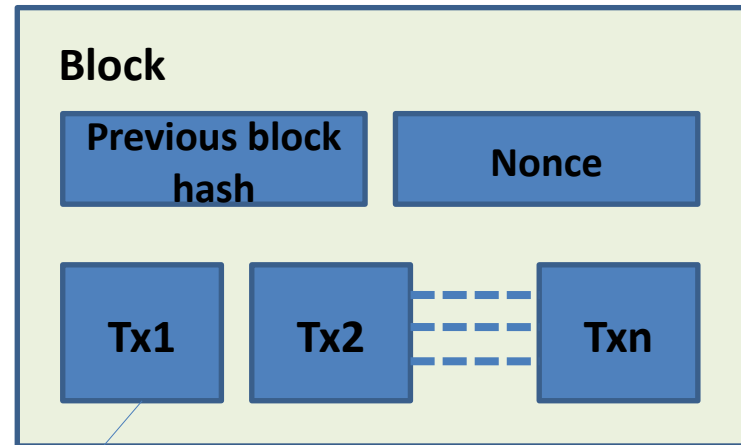


A coin is a chain of digital signatures. Each owner transfers the coin to the next owner by digitally signing a hash of the previous transaction and new owner's public key. Any verifier can verify the chain of digital signatures.

The Bitcoin Blockchain



Mining new coins



A miner has to carry out the proof of work calculation by calculating the nonce for the block. This can take trillions of iterations.

Generation transaction

1. Transaction fees
2. Coinbase reward

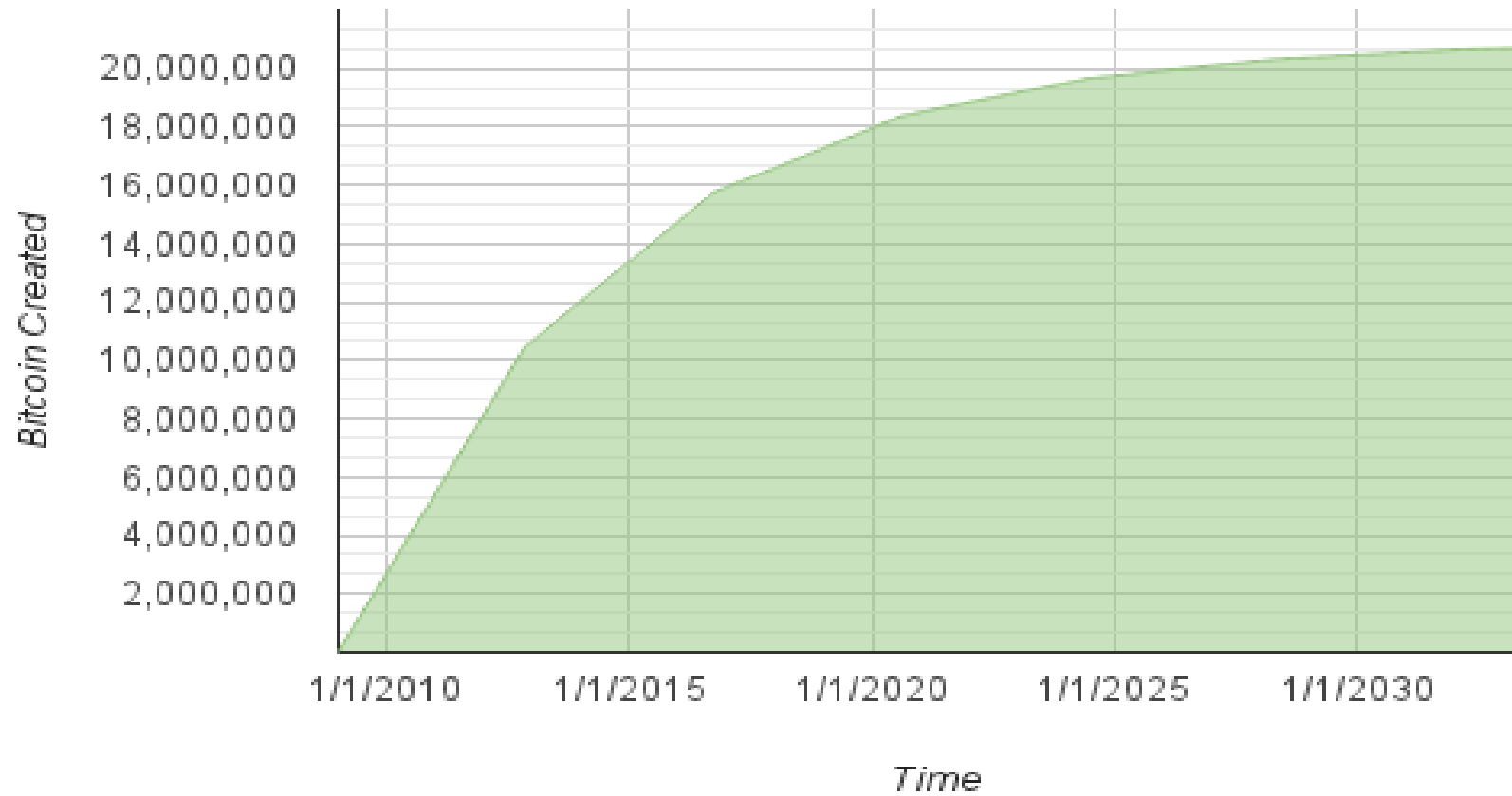
Since a block can be 1 MB at most, transactors will often offer a transaction fee as incentive to miners to include their transaction.

Is currently 6.25 bitcoin for completing a block.
It started at 50 and halves after every 210,000 blocks.

The Role of Mining

- The key innovation in Bitcoin is mining
- In a sense the term “mining” is somewhat misleading since in the real world mining for minerals is done with the primary aim to finding more of the mineral
- In the Bitcoin system one of the aims of mining is to generate new Bitcoins but it is really an incentive to miners to carry out a computation that is the basis of emergent distributed consensus. That is its real purpose. The generation of new Bitcoin is just an aligned incentive.

Bitcoin Money Supply



Independent Verification

- Once a miner competes a block it is sent out on the network.
- Each node on the network verifies the block using a set of stringent criteria. Some of these are
 - The block data structure is syntactically valid
 - The block header hash is correct (enforces the Proof-of-Work)
 - The block timestamp is less than two hours in the future (allowing for time errors)
 - The block size is within acceptable limits
 - The first transaction (and only the first) is a coinbase generation transaction
 - All transactions within the block are valid
- If a node is able to verify a block only then will it propagate it further otherwise it will reject it

Bitcoin Price History Chart (Since 2009)



Types of Blockchains

- Three common types of blockchains are
- Public blockchains
 - e.g. Bitcoin – Anyone can participate in any operation. Totally decentralized and emergent consensus
- Private blockchains
 - A single organization has sole authority over the network. Not everyone can participate. Requires trust in the entity controlling the network
- Permissioned blockchains
 - Several authorized entities can carry out operations that update blocks
- Both private and permissioned blockchains restrict block updates to one or a set of authorised entities. The blockchain may be read by anyone or there may be restrictions even on who can read it.

Applications of Blockchains

- Money Transfer and Payment Processing.
- Supply Chains Monitoring.
- Retail Programs Based on Loyalty Rewards.
- Digital IDs.
- Sharing of Data.
- Protection of Royalty and Copyright.
- Digital voting.
- Transfer of Real Estate, Land, and Auto Title.

Major Public Blockchains

- Bitcoin
 - Is strictly a cryptocurrency blockchain
- Ethereum
 - Is a platform that can be used to develop other applications.
Introduced the concept of a smart contract

Summary

- While data is widely recognized to be a very important asset, we do not have well accepted methods to value data and our current accounting practices do not reflect the value of the data assets on organizational balance sheets
- The consequences of inadequate/ambiguous/subjective data valuation techniques result in inadequate budgetary provisions to maintain and secure data assets and are a significant reason that objective and satisfactory insurance products are not available
- Data is also used to store value in the form of digital, virtual and cryptocurrencies
- The oldest practical cryptocurrency is bitcoin
- The technology of blockchains enables the maintenance of distributed ledgers which are write once only. This allows several important applications to be built on blockchain technology